

Course code : **CSE3009**
Course title : **No SQL Data Bases**
Module : **3**
Topic : **7**

Redis Commands

Objectives

This session will give the knowledge about

- Hash commands
- Admin commands

Hash Commands

HSET : used to set the field in the hash stored at key to value

Syntax: HSET KEY_NAME FIELD VALUE

> HSET user email example@gmail.com

(integer) 1

> HSET user lang English

(integer) 1

> HSET user gender Male

(integer) 1

> HGETALL user

1) "email"

2) "example@gmail.com"

3) "lang"

4) "English"

5) "gender"

6) "Male"

Hash Commands

HMSET : used to set the specified fields to their respective values in the hash stored at key.

Syntax: **HMSET KEY_NAME FIELD1 VALUE1 ...FIELDN VALUEN**

```
> HMSET user email
example@gmail.com language English
gender Male
OK
```

```
> HGETALL user
```

- 1) "email"
- 2) "example@gmail.com"
- 3) "language"
- 4) "English"
- 5) "gender"
- 6) "Male"

Hash Commands

HGET : used to get the value associated with the field in the hash stored at key.

Syntax: **HGET KEY_NAME
FIELD_NAME**

```
>HGET user email  
"example@gmail.com"  
> HGET user lang  
"English"
```

> HMGET user email language

- 1) "email"
- 2) "example@gmail.com"
- 3) "language"
- 4) "English"

Hash Commands

HMGET : used to get the values associated with the specified fields in the hash stored at key.

Syntax: **HMGET KEY_NAME
FIELD1...FIELDN**

> HMSET user email
example@gmail.com language English
gender Male
OK

> HMGET user email language
gender

1) "example@gmail.com"

2) "English"

3) "Male"

Hash Commands

HGETALL : used to get all fields and values of the hash stored at key.

Syntax: **HGETALL KEY_NAME**

```
> HMSET user email  
example@gmail.com language English  
gender Male  
OK
```

```
> HGETALL user
```

- 1) "email"
- 2) "example@gmail.com"
- 3) "lang"
- 4) "English"
- 5) "gender"
- 6) "Male"

Hash Commands

HDEL : used to remove the specified fields from the hash stored at a key and ignored the specified keys that do not exist within this hash.

Syntax: **HDEL KEY_NAME FIELD1..
FIELDN**

> HDEL user email
(integer) 1

> HDEL user lang gender
(integer) 2

Hash Commands

HEXISTS : used to check whether a hash field stored at key or not.

Syntax: **HEXISTS KEY_NAME
FIELD_NAME**

> HSET user email
example@gmail.com
(integer) 1

> HEXISTS user email
(integer) 1

> HEXISTS user xyz
(integer) 0

Hash Commands

HINCRBY : used to increment the number stored at the field in the hash stored at key by increment.

Syntax: **HINCRBY KEY_NAME
FIELD_NAME INCR_BY_NUMBER**

> HSET user visits 5
(integer) 1

> HINCRBY user visits 15
(integer) 20

> HINCRBY user visits -25
(integer) -5

Hash Commands

HINCRBYFLOAT : used to increment the specified field of a hash stored at key, and representing a floating point number, by the specified increment.

Syntax: HINCRBYFLOAT
KEY_NAME FIELD_NAME
INCR_BY_NUMBER

> HINCRBYFLOAT user height 85.5
"85.5"

> HINCRBYFLOAT user height 14.2
"99.7"

> HINCRBYFLOAT user height -12.5
"87.2"

Hash Commands

HKEYS : used to get all field names in the hash stored at key.

Syntax: HKEYS KEY_NAME
FIELD_NAME INCR_BY_NUMBER

> HMSET user name Sweaty gender
Female lang English
OK

> HKEYS user

1) "name"

2) "gender"

3) "lang"

Hash Commands

HVALS : used to get all values in the hash stored at key.

Syntax: **HVALS KEY_NAME FIELD VALUE**

> HVALS user

1) "Chima"

2) "Male"

3) "English"

> HMSET user name Chima gender
Male language English

OK

Admin Commands

Backup:

SAVE : used to create backup of current redis database.

> SAVE

OK

Restore:

CONFIG GET: To restore redis data just move redis backup file (dump.rdb) into your redis directory and start the server.

> CONFIG get dir

1) "dir"

2) "/var/lib/redis/6379"

Admin Commands

BGSAVE: The alternate command to create a Redis backup is BGSAVE. This command is used to save the DB in the background.

> BGSAVE

Background saving started

Redis with Java

```
import redis.clients.jedis.Jedis;
public class RedisJava {
    public static void main(String[] args) {
        //Connecting to Redis server on localhost
        Jedis jedis = new Jedis("localhost");
        System.out.println("Connection to server sucessfully");
        //check whether server is running or not
        System.out.println("Server is running: "+jedis.ping());
    }
}
```


Summary

This session will give the knowledge about

- Hash commands
- Admin commands
- Reference: <https://redis.io/commands/>
- <https://www.w3resource.com/redis/>